

Subject: Scalable ML & BDA

Module 1: Introduction to Scalable ML & Big Data

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Definition of ML:

- Machine Learning (ML) is a subfield of Artificial Intelligence (AI) that focuses on building algorithms and models that enable computers to learn from data and improve with experience without explicit programming for every task.
- In simple words, Machine Learning teaches systems to learn patterns and make decisions like humans by analyzing and learning from data.
- ML systems improve their performance by identifying patterns and making predictions based on experience.
- The main goal of machine learning is to build intelligent systems that can learn, adapt, and improve over time.

Factor	Traditional Programming	Machine Learning
Instruction Method	Explicit rules and logic	Learns patterns from data
Data Handling	Structured, predictable data	Large, often unstructured data
Outcome Predictability	Always the same result for same input	Predictions may vary as model adapts to new data
Flexibility	Limited to predefined conditions	Adjusts based on new data (self-improving)

Factor	Traditional Programming	Machine Learning
Development Process	Linear: write, debug, deploy	Iterative: train, evaluate, tune, retrain
Transparency	Easy to trace and debug	Can be opaque, often needs explainable AI
Problem Complexity	Best for simple, well-defined tasks	Best for complex, data-rich tasks

Need for Machine Learning:

Machine Learning is important because traditional programming cannot handle complex tasks or large amounts of data efficiently. ML overcomes this by learning from data and making predictions without fixed rules. It is needed for the following reasons:

1. Solving Complex Business Problems

Traditional programming struggles with tasks like language understanding and medical diagnosis. ML learns from data and predicts outcomes easily.

Examples:

- Image and speech recognition in healthcare.
- Language translation and sentiment analysis.

2. Handling Large Volumes of Data

The internet generates huge amounts of data every day. Machine Learning processes and analyzes this data quickly by providing valuable insights and real time predictions.

Examples:

- Fraud detection in financial transactions.

- Personalized feed recommendations on Facebook and Instagram from billions of interactions.

3. Automate Repetitive Tasks

ML automates time consuming, repetitive tasks with high accuracy hence reducing manual work and errors.

Examples:

- Gmail filtering spam emails automatically.
- Chatbots handling order tracking and password resets.
- Automating large scale invoice analysis for key insights.

4. Personalized User Experience

ML enhances user experience by tailoring recommendations to individual preferences. It analyze user behavior to deliver highly relevant content.

Examples:

- Netflix suggesting movies and TV shows based on our viewing history.
- E-commerce sites recommending products we're likely to buy.

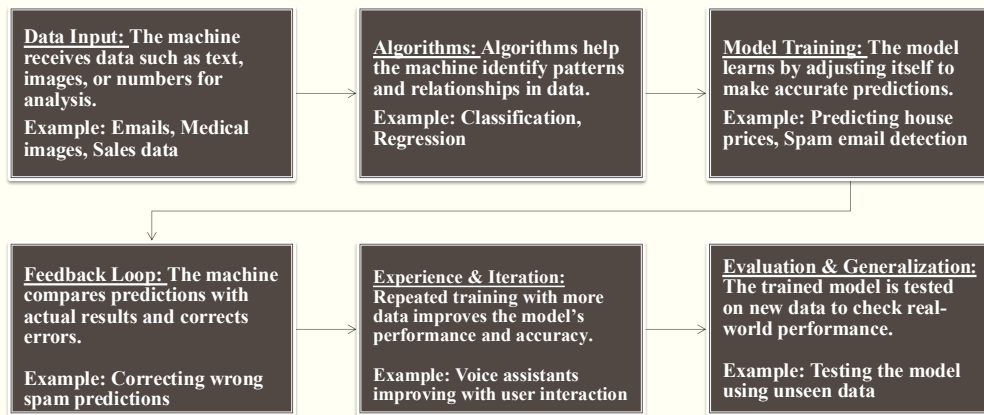
5. Self Improvement in Performance

ML models evolve and improve with more data helps in making them smarter over time. They adapt to user behavior and increase their performance.

Examples:

- Voice assistants like Siri and Alexa learning our preferences and accents.
- Search engines refining results based on user interaction.
- Self driving cars improving decisions using millions of miles of driving data.

How Machines Learn from Data



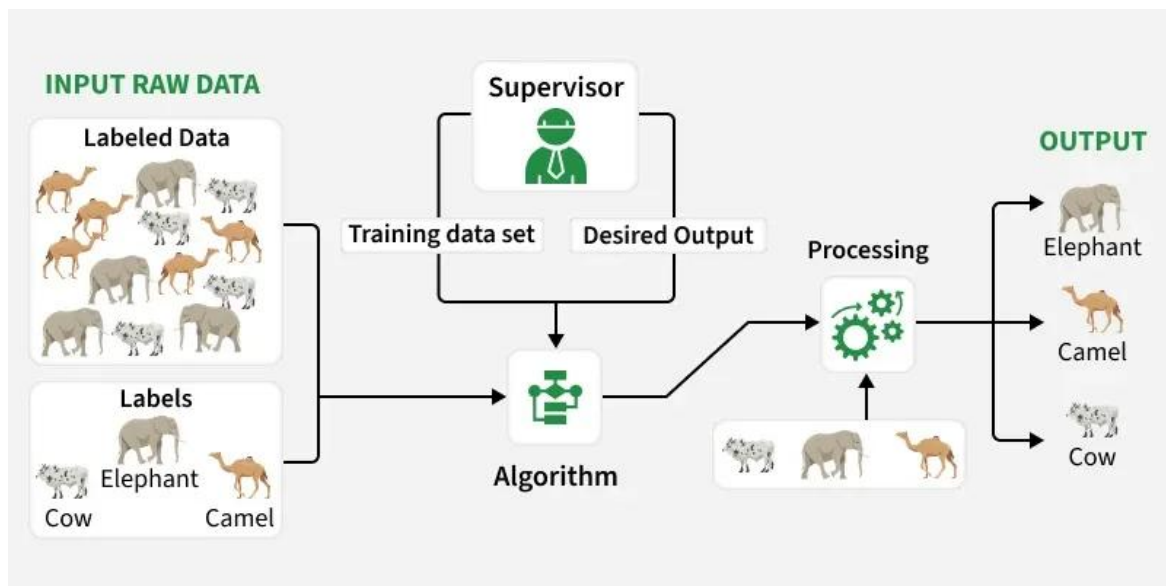
Importance of Data in Machine Learning

1. Data is the foundation of machine learning (ML). Without good quality data, models cannot learn, perform well or make accurate predictions.
2. Data provides examples for models to learn patterns and relationships.
3. High-quality and diverse data improves accuracy and generalization.
4. Proper feature selection enhances model performance.
5. Separate validation and test data ensure reliable evaluation.
6. Continuous data collection helps improve models over time.

Types Of Machine Learning

1. Supervised Machine Learning

It is defined as when a model gets trained on a "Labeled Dataset". Labelled datasets have both input and output parameters. In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.



Example: If you train a model using labeled images of cats and dogs, it learns the features of each. When shown a new image, it predicts whether it's a cat or a dog.

There are two main categories of supervised learning that are mentioned below:

- a. **Classification:** It predicts categorical outputs, meaning it assigns data into predefined classes like spam/non-spam emails or disease risk categories. These algorithms learn to map input features to discrete labels. Here are some classification algorithms: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Naive Bayes, Support Vector Machine (SVM)
- b. **Regression:** It predicts continuous values, such as house prices or product sales. It learns the relationship between input features and a numerical target variable. Here are some regression algorithms: Linear Regression, Polynomial Regression, Ridge Regression, Lasso Regression, Decision Tree, Random Forest.

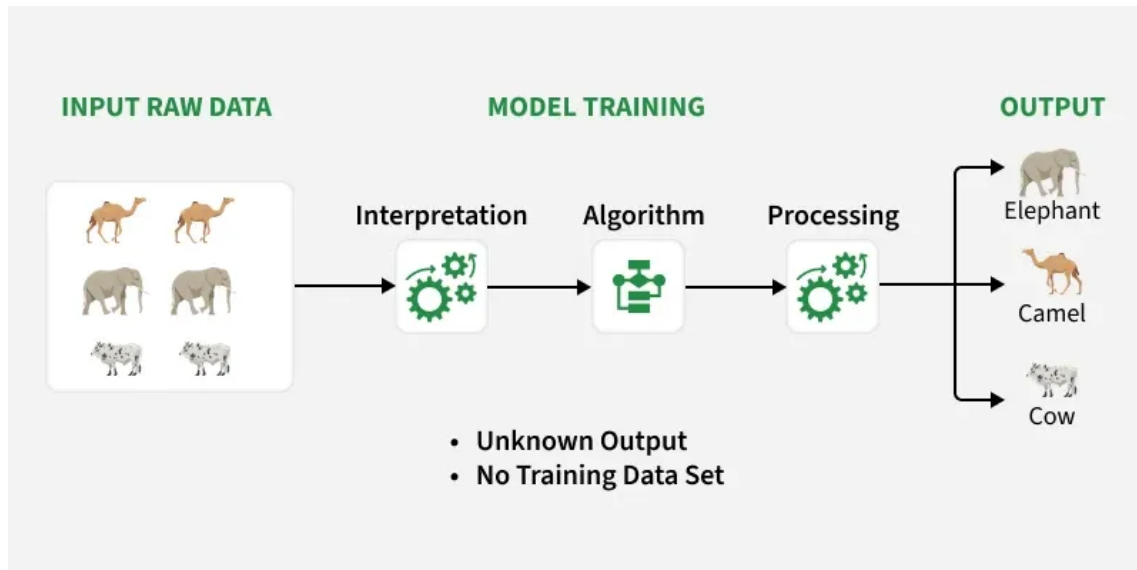
Where to Use Supervised Learning

- When you have labeled data and want to predict outcomes.
- Ideal for classification (like spam detection) or regression tasks (like price forecasting).
- Best used in domains where historical data with outcomes is already available.

Applications: Image, Speech and Text Processing, Predictive Analytics, Recommendation and Personalization, Healthcare and Finance, Automation and Control.

2. Unsupervised Machine Learning

Unsupervised Learning works with unlabeled data, meaning there are no predefined outputs. The algorithm finds hidden patterns, groups or relationships within the data on its own. It's mainly used for clustering, dimensionality reduction and data visualization.



Example: If you have customer data without labels, the algorithm can group similar customers based on purchase behavior useful for segmentation and marketing.

There are two main categories of unsupervised learning that are mentioned below:

- Clustering:** Clustering is the process of grouping data points into clusters based on their similarity. This technique is useful for identifying patterns and relationships in data without the need for labeled examples. Common techniques include: K-Means, DBSCAN, Mean-Shift.
- Dimensionality Reduction Techniques:** It helps reduce the number of features while preserving important information. Common techniques include: Principal Component Analysis (PCA), Independent Component Analysis (ICA).

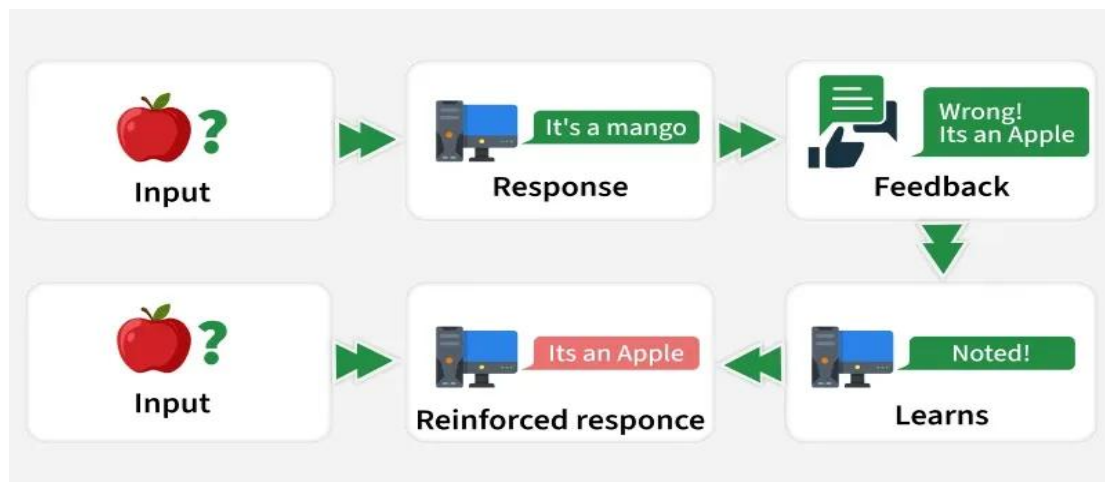
Where to Use Unsupervised Learning

- When data is unlabeled or unstructured.
- Useful for exploratory analysis, clustering or feature extraction.
- Common in marketing, recommendation systems and fraud detection where patterns matter more than labels.

Applications: Clustering and Segmentation, Anomaly Detection, Dimensionality Reduction, Recommendation and Marketing, Data Preprocessing and Analysis.

3. Reinforcement Learning:

It trains an agent to make a sequence of decisions through trial and error. The agent interacts with the environment, receives feedback in the form of rewards or penalties and learns optimal actions over time.



Example: An AI agent learning to play chess gets positive feedback for good moves and negative for poor ones. Over time, it learns strategies to win more often.

Types of Reinforcement Learning

- **Positive Reinforcement:** Rewards desired behavior (e.g., giving points for correct answers).
- **Negative Reinforcement:** Removes negative outcomes to encourage good actions (e.g., turning off a buzzer after the right move).

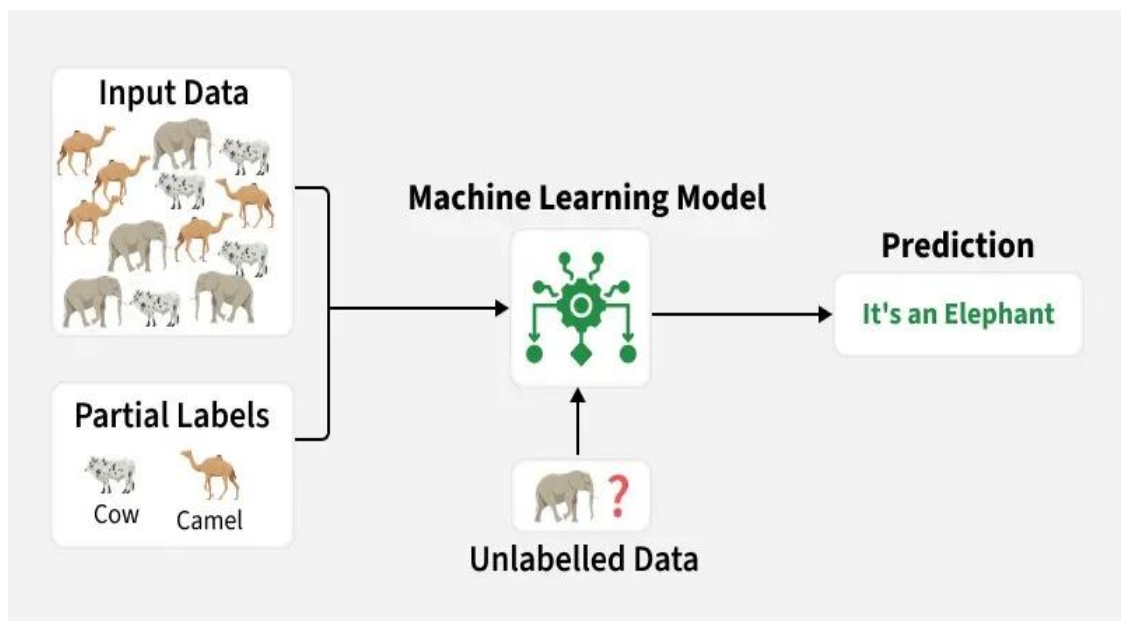
Where to Use Reinforcement Learning

- When you need an agent to learn by interacting with an environment.
- Best for decision-making or optimization tasks involving trial and feedback loops.
- Used when long-term performance or adaptive behavior is more important than immediate accuracy.

Applications: Gaming and Simulation, Robotics and Automation, Autonomous Vehicles, Healthcare and Finance, Recommendation and Personalization, Industrial and Energy Management.

4. Semi-Supervised Learning:

Semi-Supervised Learning combines both Supervised and Unsupervised approaches. It uses a small set of labeled data and a large set of unlabeled data for training useful when labeling is costly or time-consuming.



Example: Consider that we are building a language translation model, having labeled translations for every sentence pair can be resource intensive. It allows the models to learn from labeled and unlabeled sentence pairs, making them more accurate. This technique has led to significant improvements in the quality of machine translation services.

Where to Use Semi-Supervised Learning

- When you have limited labeled data but plenty of unlabeled data.
- Useful for domains with high labeling costs, such as medical, NLP or image datasets.
- Ideal when unlabeled data still holds valuable information that can improve learning performance.

Applications: Limited Labeled Data, High-Cost Labeling, Valuable Unlabeled Data.

5. Self-Supervised Learning:

Self-Supervised Learning is a modern approach where models generate their own labels from raw data. It doesn't rely on manual annotation instead, the model learns by predicting parts of data from other parts.

Where to Use Self-Supervised Learning?

- When manual labeling is impossible or expensive.
- Suitable for large-scale datasets like text, audio and images.
- Best for pre-training models that can later be fine-tuned for specific supervised tasks.

Scalable ML Definition:

Scalable Machine Learning refers to techniques that allows ML models to efficiently handle increasing amounts of data & computational workload, often by distributing data & computations across multiple machines.

Challenges of Scalable ML:

- i) Large Data Volume
- ii) High Training Time
- iii) Velocity
- iv) Scalability
- v) Variety
- vi) Computational Resource Limitations

1. Volume - Scale of Data

The name 'Big Data' itself is related to a size which is enormous. Volume is a huge amount of data. To determine the value of data, size of data plays a very crucial role. If the volume of data is very large, then it is actually considered as a 'Big Data'. This means whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data. Hence while dealing with Big Data it is necessary to consider a characteristic 'Volume'.

2. Velocity - Speed of Data

Velocity refers to the high speed of accumulation of data. In Big Data velocity data flows in from sources like machines, networks, social media, mobile phones etc. There is a massive and continuous flow of data. This determines the potential of data that how fast the data is generated and processed to meet the demands. Sampling data can help in dealing with the issue like 'velocity'.

3. Variety - Diversity of Data

It refers to nature of data that is structured, semi-structured and unstructured data. It also refers to heterogeneous sources. Variety is basically the arrival of data from new sources that are both inside and outside of an enterprise. It can be structured, semi-structured and unstructured.

- **Structured data:** This data is basically an organized data. It generally refers to data that has defined the length and format of data.
- **Semi- Structured data:** This data is basically a semi-organised data. It is generally a form of data that do not conform to the formal structure of data. Log files are the examples of this type of data.
- **Unstructured data:** This data basically refers to unorganized data. It generally refers to data that doesn't fit neatly into the traditional row and column structure of the relational database. Texts, pictures, videos etc. are the examples of unstructured data which can't be stored in the form of rows and columns.

4. High Training Time: As the dataset size & model complexity increase, the time required to train ML models also increases significantly.

5. Scalability: Scalability is the ability of a machine learning system to maintain performance as the amount of data, users, or computational workload increases.

6. Limited Computational Resources: Large ML models require high CPU performance, GPUs, memory, and storage.

Traditional Machine Learning: It refers to training & processing machine learning models on a single computer or server where all the data & computations are handled by one machine.

Distributed Machine Learning: It is an approach in which data & computations are divided among multiple computers(nodes) that work together to train a machine learning model more efficiently.

Big Data:

1. Big Data refers to vast and rapidly growing volumes of data that are too large and complex for traditional data processing tools to manage. This data comes in many forms structured (e.g., tables), semi-structured (e.g., JSON, XML), and unstructured (e.g., text, images, video).
2. With the explosion of devices, sensors, online services, and digital platforms, data is now generated at an unprecedented rate. This growth makes it essential for organizations to adopt advanced tools and technologies to capture, store, analyze, and utilize this data effectively.

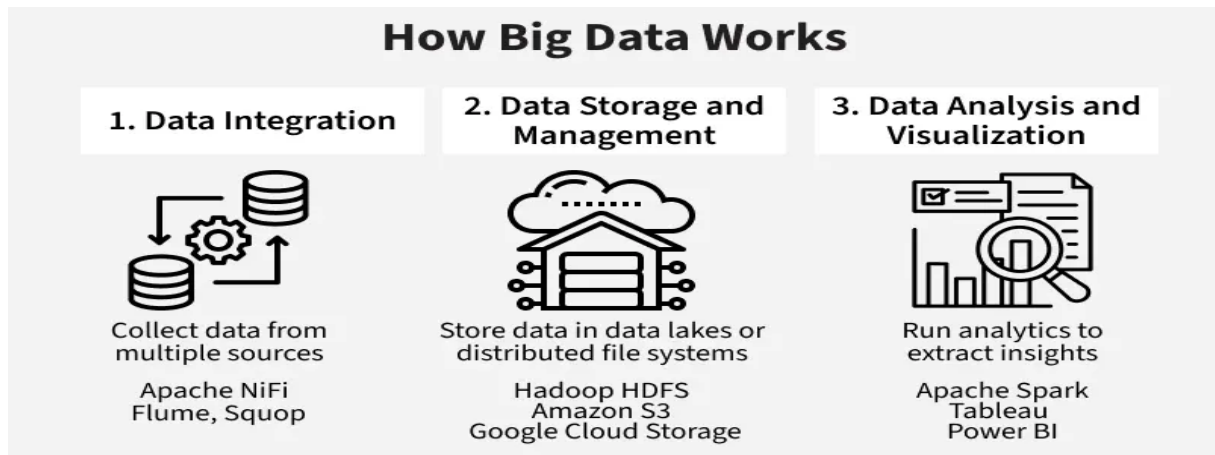
Practical Uses of Big Data

Organizations use Big Data to:

- Make smarter decisions by identifying trends and patterns
- Predict customer behavior and personalize user experiences
- Improve operational efficiency by finding process inefficiencies
- Innovate faster by identifying new business opportunities
- Enhance risk management by detecting fraud or security threats

Big Data transforms raw information into actionable insights that help companies gain a competitive edge.

Types of Big Data: Data → Structured, Semi-Structured, Unstructured



Big Data Technologies – Hadoop:

Hadoop is an open-source framework for storing and processing large-scale data across distributed clusters using commodity hardware. The Hadoop Ecosystem is a suite of tools and technologies built around Hadoop's core components (HDFS, YARN, MapReduce and Hadoop Common) to enhance its capabilities in data storage, processing, analysis and management.

Components of Hadoop Ecosystem:

Hadoop Ecosystem comprises several components that work together for efficient big data storage and processing:

- **MapReduce:** A programming model for batch data processing.
- **Spark:** Provides fast, in-memory data processing.
- **HBase:** A NoSQL database for real-time read/write access.
- **Mahout & Spark MLlib:** Libraries for scalable machine learning.
- **Solr & Lucene:** Tools for full-text search and indexing.
- **Zookeeper:** Manages coordination and configuration across the cluster.
- **Oozie:** A workflow scheduler for managing Hadoop jobs.

HDFS

HDFS is a core component of Hadoop ecosystem, designed to store large volumes of structured or unstructured data across multiple nodes. It manages metadata through log files and splits storage tasks between two main parts:

- **NameNode (master):** Stores metadata (data about data) and requires fewer resources.
- **DataNodes (slaves):** Store actual data on commodity hardware, making Hadoop cost-effective.

HDFS handles coordination between clusters and hardware, serving as the backbone of entire Hadoop system.

YARN

YARN (Yet Another Resource Negotiator) is resource management layer of Hadoop, responsible for scheduling and allocating resources across the cluster. It has three key components:

- **ResourceManager:** Allocates resources to various applications in the system.
- **NodeManager:** Manages resources (CPU, memory, etc.) on individual nodes and reports to ResourceManager.
- **ApplicationMaster:** Acts as a bridge between ResourceManager and NodeManager, handling resource negotiation for each application.

Together, they ensure efficient resource utilization and smooth execution of jobs in the Hadoop cluster.

MapReduce

MapReduce enables distributed and parallel data processing on large datasets. It allows developers to write programs that transform big data into manageable results.

- **Map():** Processes input data by filtering, sorting and organizing it into key-value pairs.
- **Reduce():** Takes the output from Map(), aggregates the data and summarizes it into a smaller, consolidated set of results.

Together, they efficiently handle large-scale data transformations across the Hadoop cluster.

PIG

Pig is a platform developed by Yahoo for analyzing large datasets using Pig Latin, a SQL-like scripting language designed for data processing.

- It simplifies complex data flows and handles MapReduce operations internally.

- The processed results are stored in HDFS.
- Pig Latin runs on Pig Runtime, similar to Java on JVM.
- It enhances programming ease and optimization, making it a key part of Hadoop ecosystem.

HIVE

Hive uses a SQL-like interface (HQL: Hive Query Language) to read and write large datasets.

- It supports both real-time and batch processing, making it highly scalable.
- Hive supports all standard SQL datatypes, easing query operations.

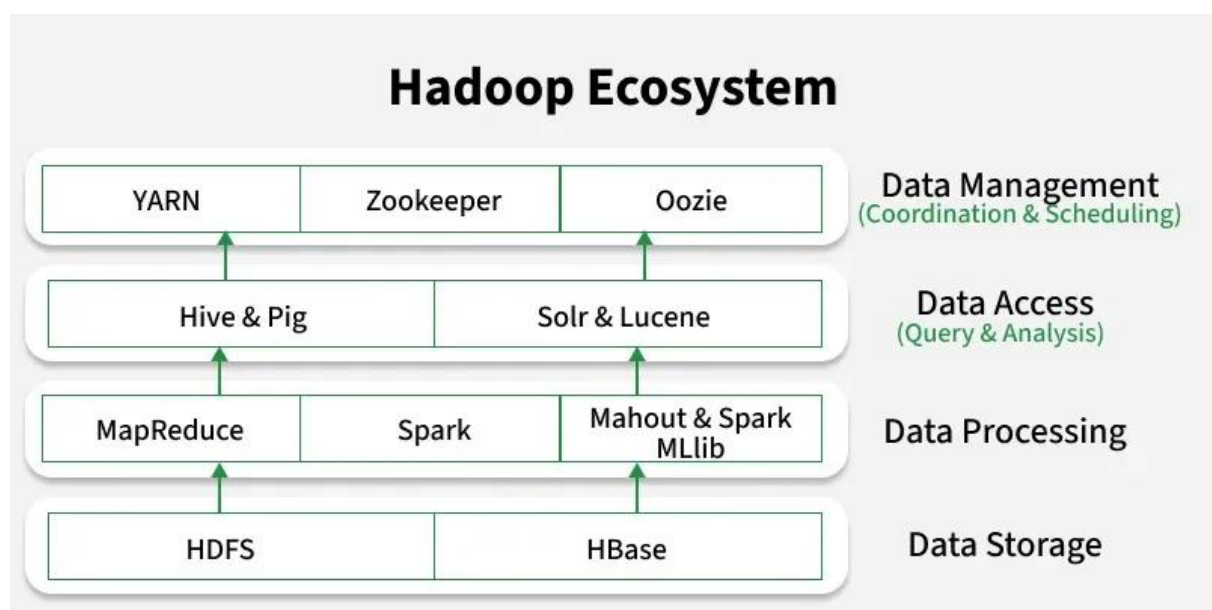
It has two main components:

- **JDBC/ODBC drivers:** manage data connections and access permissions.
- **Hive Command Line:** used for query execution and processing.

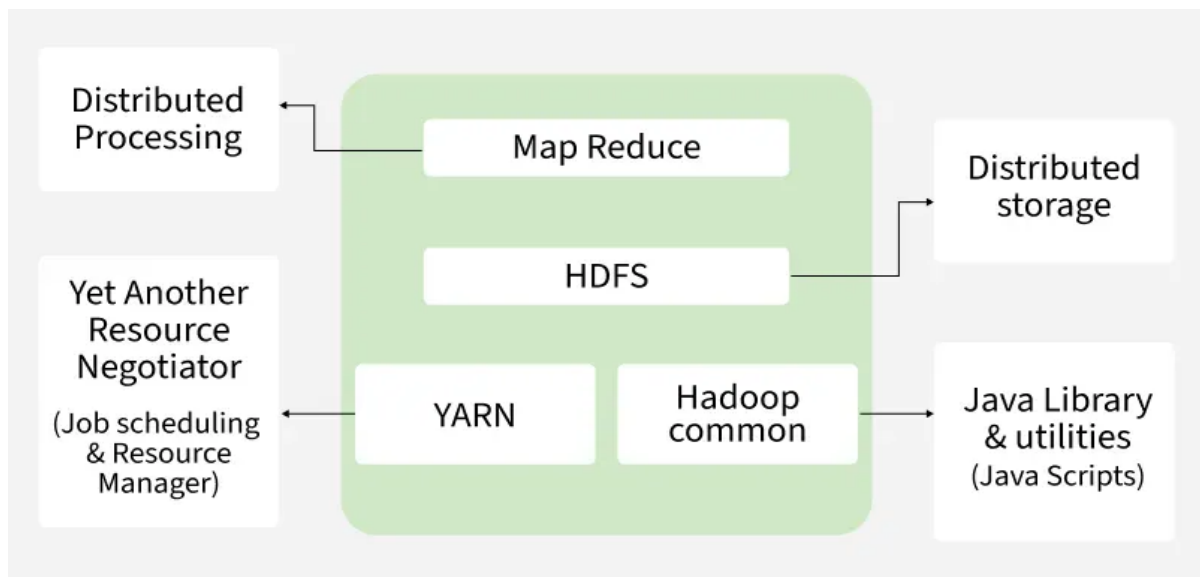
Mahout

Mahout brings machine learning capabilities to Hadoop-based systems by enabling applications to learn from data using patterns and algorithms.

- It provides built-in libraries for clustering, classification and collaborative filtering.
- Users can invoke algorithms as needed through Mahout's scalable, distributed libraries.



Hadoop Architecture:



Advantage of Hadoop:

1. Cost effective.
2. Processing operation is done at a faster speed.
3. Best to be applied when a company is having a data diversity to be processed.
4. Creates multiple copies.
5. Saves time and can derive data from any form of data.

Disadvantage of Hadoop:

1. Can't perform in small data environments
2. Built entirely on java
3. Lack of preventive measures
4. Potential stability issues
5. Not fit for small data

Apache Spark:

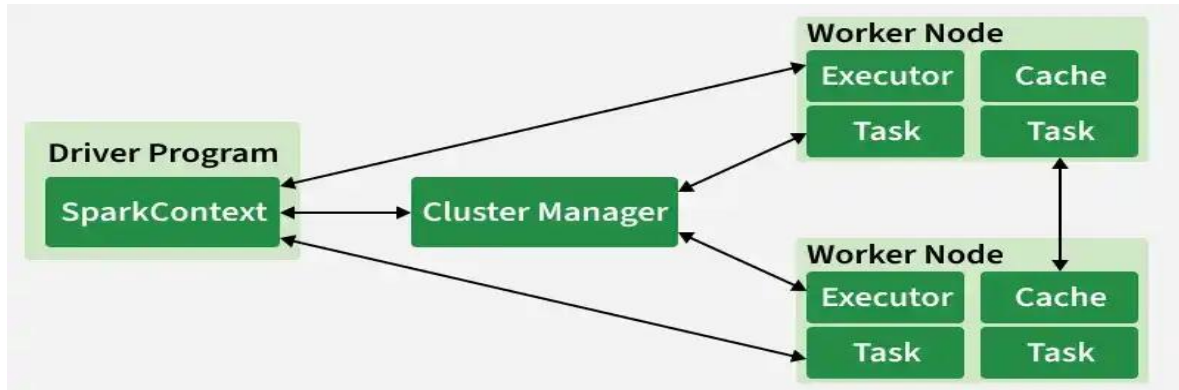
Apache Spark is an open-source distributed computing framework built for large-scale data processing and analytics.

It processes massive datasets across clusters of machines with high speed and reliability.

- It is designed for large-scale distributed data processing.
- Handles massive datasets (terabytes to petabytes).

- Uses in-memory computation for high performance.
- Provides fault tolerance and horizontal scalability.

Core Architecture:



- Driver Program - Controls the Spark application and coordinates the execution.
- SparkContext - Connects the Driver Program with the cluster and coordinates Spark jobs.
- Cluster Manager - Allocates resources to the Spark application.
- Worker Nodes - Machines that provide resources for processing.
- Executor - Runs tasks on worker nodes and can cache data in memory.
- Task - A unit of work performed on a partition of data.

Key abstractions:

- Resilient Distributed Datasets (RDDs) - Low-level, immutable, partitioned collections with lineage for fault recovery (still foundational but less used directly now).
- DataFrames / Datasets - Higher-level, structured (with schema), optimized via Catalyst optimizer and Tungsten execution engine (columnar, codegen).
- Directed Acyclic Graph (DAG) - Execution plan: Spark builds a DAG of stages (shuffle boundaries separate them).

Built-in libraries (unified engine):

- Spark SQL - Structured data, ANSI SQL, DataFrames.

- Spark Streaming / Structured Streaming - Real-time (micro-batch or continuous/real-time mode in 4.x for ms latency).
- MLlib - Scalable machine learning.
- GraphX - Graph computation

Why is Apache Spark Fast?

- In-Memory Processing: Keeps frequently used data in memory for faster access.
- Parallel Processing: Processes data across multiple machines simultaneously.
- Caching: Reuses data stored in memory for repeated operations.
- Fault Tolerance: Can recover lost data through lineage.

Key Use Cases (Real-World in 2026):

Spark powers critical systems across industries:

- **ETL / Data Pipelines** - Ingest, clean, transform, load massive data (bronze → silver → gold layers).
- **Real-Time Analytics & Streaming** - Fraud detection, live dashboards, IoT sensor processing.
- **Machine Learning & AI** - Train models on petabyte-scale data (recommendations, predictive maintenance).
- **Recommendation Engines** - Netflix/Amazon personalize content/products using collaborative filtering.
- **Log & Event Processing** - Analyze application logs for monitoring/anomalies.
- **Fraud & Risk Detection** - Finance/security combine transaction data with external signals.
- **Healthcare** - Real-time patient monitoring, population health analytics.

Companies like Netflix (recommendations + log analytics), Amazon (order fulfillment), banks (fraud), and OTT platforms rely on Spark at massive scale.

Advantage of Spark:

1. Perfect for interactive processing, iterative processing and event stream processing

2. Flexible and powerful
3. Supports for sophisticated analytics
4. Executes batch processing jobs faster than MapReduce
5. Run on Hadoop alongside other tools in the Hadoop ecosystem

Disadvantage of Spark:

1. Consumes a lot of memory
2. Issues with small file
3. Less number of algorithms
4. Higher latency compared to Apache fling

Apache Flink

In the current generation, Apache Flink is the big giant tool that is nothing but 4G of Big Data. It's the true stream processing framework. Flink's kernel (core) is a streaming runtime that provides distributed processing, fault tolerance. Flink processes events at a constantly high speed with low latency. It schemes the data at lightning-fast speed. Apache Flink is the large-scale data processing framework that we can reuse when data is generated at high velocity. This is an important open-source platform that can address numerous types of conditions efficiently:

1. Batch Processing
2. Iterative Processing
3. Real-time stream processing
4. Interactive processing
5. In-memory processing
6. Graph Processing

Flink is the volition to MapReduce, it processes data further than 100 times faster than MapReduce. It's independent of Hadoop but it can use HDFS (Hadoop Distributed File System)

to read, write, store, process the data. Flink doesn't give its own data storehouse system. It takes data from the distributed storehouse.

Architecture of Apache Flink:

Apache Flink is a distributed stream-processing and batch-processing framework. Its architecture mainly consists of JobManager and TaskManagers.

1. JobManager

- It is the master/coordinator of the Flink cluster.
- It receives jobs submitted by the client.
- It converts the job into executable tasks and schedules them.
- It coordinates and monitors the execution of the job.
- It manages the overall execution of the Flink job.

2. TaskManager

- It is the worker node of the Flink cluster.
- It executes the tasks assigned by the JobManager.
- It provides the computational resources required for processing.
- Multiple TaskManagers can work together to process a job in parallel.

Execution Model of Apache Flink:

The execution of a Flink program takes place in the following steps:

1. Program

- The user writes a Flink program.
- The program defines the required data processing operations.

2. Parse and Optimize

- Flink analyzes and processes the program.
- It performs tasks such as parsing, type checking/type inference, and optimization.

3. DataFlow Graph

- The Flink program is converted into a DataFlow Graph.
- The graph represents the sequence of data-processing operations.
- It consists of operators and the flow of data between them.

4. JobManager

- The JobManager receives the job.
- It schedules the tasks on available TaskManagers.
- It coordinates and monitors job execution.
- It manages information related to the execution of the job.

5. TaskManager

- TaskManagers execute the assigned tasks.
- They perform the actual data processing.
- Multiple TaskManagers can execute different parts of the job in parallel.

Why Apache Flink?

- Real-Time Stream Processing: Processes continuously arriving data.
- Low Latency: Produces results quickly as data arrives.
- High Performance: Processes large volumes of data efficiently.
- Scalability: Can process data across multiple machines.
- Fault Tolerance: Can continue processing despite failures.

Differences Between Hadoop, Spark and Flink

Based On	Apache Hadoop	Apache Spark	Apache Flink
Data Processing	Designed for batch processing.	Supports batch processing and stream processing.	Supports both batch and stream processing in a single runtime.

Based On	Apache Hadoop	Apache Spark	Apache Flink
Data Flow	Supports linear data flow and does not contain any loops.	Supports cyclic data flow and represented as (DAG) direct acyclic graph.	Uses a controlled cyclic graph at runtime, which efficiently supports ML algorithms.
Computation Model	Supports batch-oriented model.	Supports micro-batching computational model.	Supports continuous operator-based streaming model.
Performance	Slowest	Faster than Hadoop	Fastest
Memory management	Configurable Memory supports both dynamically or statically management.	Automatic memory management.	Automatic memory management
Fault tolerance	Highly fault-tolerant using a replication mechanism.	Provides fault tolerance through lineage.	Distributed snapshots results in high throughput.
Scalability	Highly scalable.	Highly scalable.	Highly scalable.
Iterative Processing	Does not support Iterative Processing.	Supports Iterative Processing.	Supports Iterative Processing and iterate data with streaming architecture.

Based On	Apache Hadoop	Apache Spark	Apache Flink
Supported Languages	Java, C++, Python, Ruby, etc.	Java, Python, R, Scala.	Java, Python, R, Scala.
Cost	Uses commodity hardware so Less expensive.	Need more RAM so Cost is relatively High.	Also need lots of RAM so Cost is relatively High.
SQL support	Users can run SQL queries using Apache Hive.	Users can run SQL queries using Spark SQL and Hive.	Supports Table-API similar to SQL expression also SQL(planned expansion).
Caching	Not Supported.	Supported (in memory)	Supported (in memory)
Machine Learning	Apache Mahout is used for ML.	Spark is used for implementing ML algorithms with its own ML libraries.	FlinkML library is used for ML implementation.
Backpressure Handling	Handles through Manual Configuration.	Handles through Manual Configuration.	Handles Implicitly through System Architecture
Criteria for Windows	Does not have windows criteria since it does not support streaming.	Spark has time-based window criteria.	Flink has record-based Window criteria.

Based On	Apache Hadoop	Apache Spark	Apache Flink
Apache License	Apache License 2.	Apache License 2.	Apache License 2.

What is Data Ingestion?

Data ingestion refers to the process of importing, transferring, or loading data from various external sources into a system or storage infrastructure where it can be stored, processed, and analyzed. It's a foundational step in the data pipeline, especially in data-driven organizations where large volumes of data are generated and collected from different sources.

Data ingestion is a critical process in modern data architectures, especially in big data and data analytics environments, as it lays the foundation for subsequent data processing, analysis, and decision-making. Efficient data ingestion ensures that organizations can leverage their data assets effectively to gain insights, drive innovation, and make data-driven decisions.

Importance of Data Ingestion:

1. Better Data Quality

- Data from multiple sources can be combined and checked.
- Errors, missing values, and inconsistencies can be identified and corrected.

2. Improved Decision Making

- Organizations can analyze data from different sources together.
- It helps identify patterns, trends, and useful insights for better decisions.

3. Automation of Business Processes

- Data collection and integration can be automated.
- This reduces manual effort, time, and operational costs.

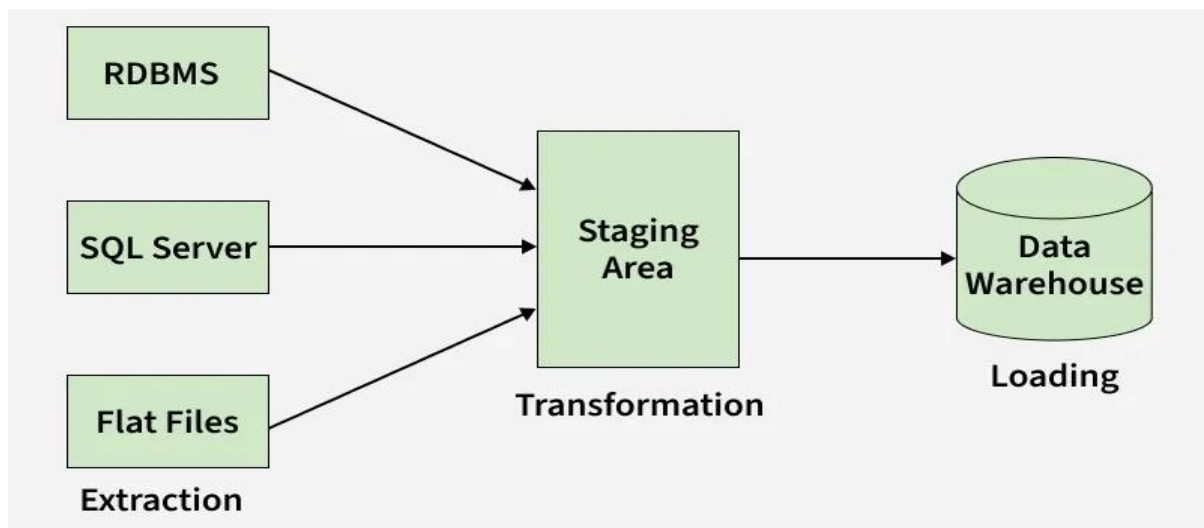
4. Centralized Data Access

- Data from different sources can be brought into a common system.
- This makes data easier to access, manage, and analyze.

ETL:

ETL (Extract, Transform, Load) is a data integration process used to collect data from multiple sources, transform it into a consistent format and load it into a target system such as a data warehouse or data lake. It helps organizations organize and prepare data for analysis, reporting and decision making.

- Extracts data from different sources and systems.
- Transforms data by cleaning, validating and organizing it.
- Loads the processed data into a database, data warehouse or data lake.
- Ensures data is accurate, consistent and ready for business use.



Need for Scalable ETL in Big Data

When the amount of data becomes very large, such as millions or billions of records from multiple sources, processing it on a single machine may be slow and inefficient.

Therefore, Big Data systems require scalable ETL pipelines that can handle large volumes of data efficiently using distributed processing.

Working of ETL

ETL works in three main stages: Extract, Transform and Load. These stages help collect data from different sources, prepare it for analysis and store it in a target system.

1. Extract

The extraction stage involves collecting raw data from various sources and moving it to a temporary storage area called the staging area. The data may come in different formats and structures. Common Data Sources:

- SQL and NoSQL databases
- CRM and ERP systems
- JSON and XML files
- Flat files (CSV, TXT, etc.)
- Emails
- Web pages

2. Transform

In the transformation stage, the raw data is cleaned and processed to make it suitable for analysis and storage. Common Transformation Tasks:

- Filtering and cleaning data
- Combining data from multiple sources
- Removing duplicate records
- Performing calculations and aggregations
- Validating data quality and compliance
- Encrypting or securing sensitive information
- Formatting data to match the target system structure

3. Load

In the loading stage, the transformed data is transferred to a target system such as a data warehouse, data lake or database. Data can be loaded all at once, incrementally or through periodic refreshes.

- Stores processed data in the target system.
- Supports full, incremental or refresh based loading.

- Makes data available for reporting, analytics and machine learning.
- Often scheduled during low activity periods to reduce system impact.

Advantages

- Improves data quality by removing errors, inconsistencies and duplicate records.
- Automates data integration processes, reducing manual effort and saving time.
- Organizes and prepares data for reporting, analytics and business intelligence.
- Converts raw data into a structured format suitable for analysis.
- Efficiently handles large volumes of data from multiple sources.

Limitations

- Can become complex when dealing with multiple data sources and formats.
- Requires regular maintenance as data sources and business requirements evolve.
- May lead to data loss or inconsistencies if transformations are not handled properly.
- Large datasets and complex transformations can introduce processing delays.
- Performance and scalability may become challenging as data volume grows.